

pico-JEPA v2: When Do Many Small Models Beat a Single Large One?

Heterogeneous ensembles and stacking for label-free video learning on an affordable PC

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github.com/BHI-Research/pico-jepa-v2

Understanding video is expensive: labels, data, and compute

- ▶ Labeling video by hand is slow and costly.
- ▶ Self-supervised learning learns by watching unlabeled videos.
- ▶ But reference models are trained at industrial scale.
- ▶ Our setting: **a desktop PC with a 6 GB GTX 1060.**

DATA

≈ 670×

fewer pre-training
videos than V-JEPA

PARAMETERS

≈ 300×

fewer than the
largest V-JEPA 2
encoder

HARDWARE

6 GB

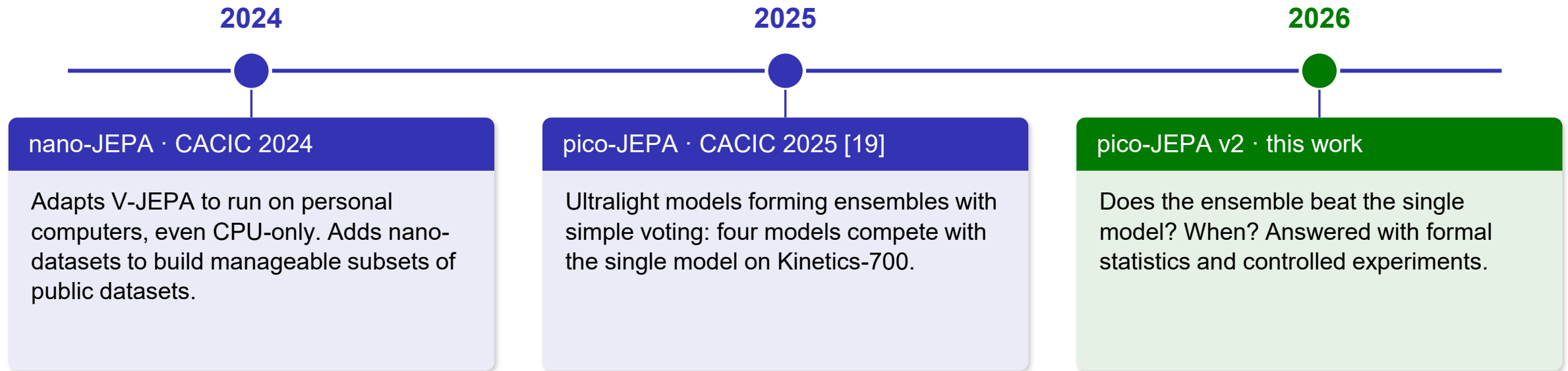
of video memory: a
single desktop GPU

Central question

With fixed compute and data, is it better to train a single model on everything... or several small models, each on a different portion, that combine their predictions?

“Strength in numbers” — but not always: the interesting question is when.

Background: from nano-JEPA to pico-JEPA v2



What v2 contributes

- 1 Heterogeneous ensemble**
each submodel has a different head
- 2 Stacking**
a meta-model learns whom to listen to
- 3 Formal statistics**
bootstrap, 95% CI, and robustness check
- 4 Map of conditions**
when the advantage appears and when it vanishes

From “competes with the single model” to “beats it, and we know when.”

State of the art I: self-supervised learning with JEPA

What does JEPA propose?

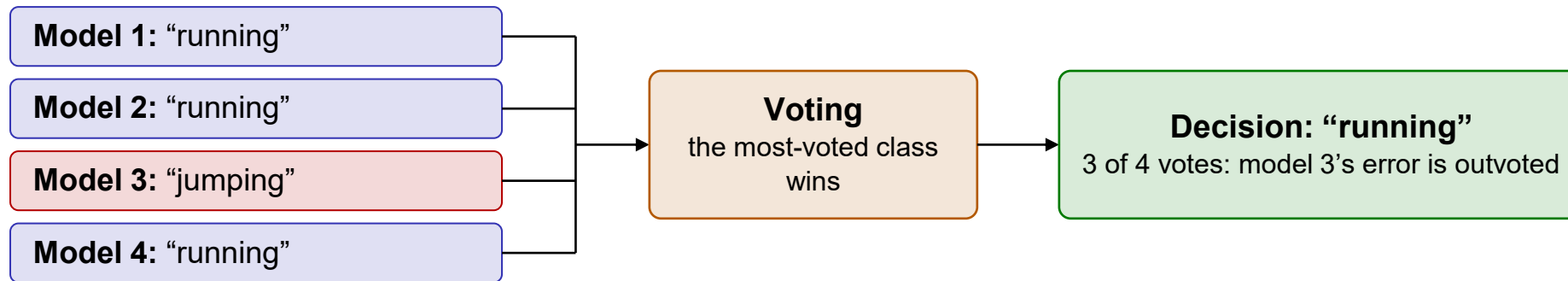
- ▶ JEPA learns **without labels**, by predicting representations of hidden parts of the data.
- ▶ Unlike reconstruction methods, **it does not try to recover pixels**, but relevant information in representation space.
- ▶ In video, **V-JEPA** extends this idea by learning spatial and temporal relations between different regions of a sequence.
- ▶ The goal is to obtain **semantic, useful representations**, without spending capacity on irrelevant visual details.

Why is it interesting for our research?

By working directly on representations and avoiding pixel reconstruction, JEPA offers an attractive foundation for building more efficient and compact self-supervised models.

State of the art II: the wisdom of model crowds

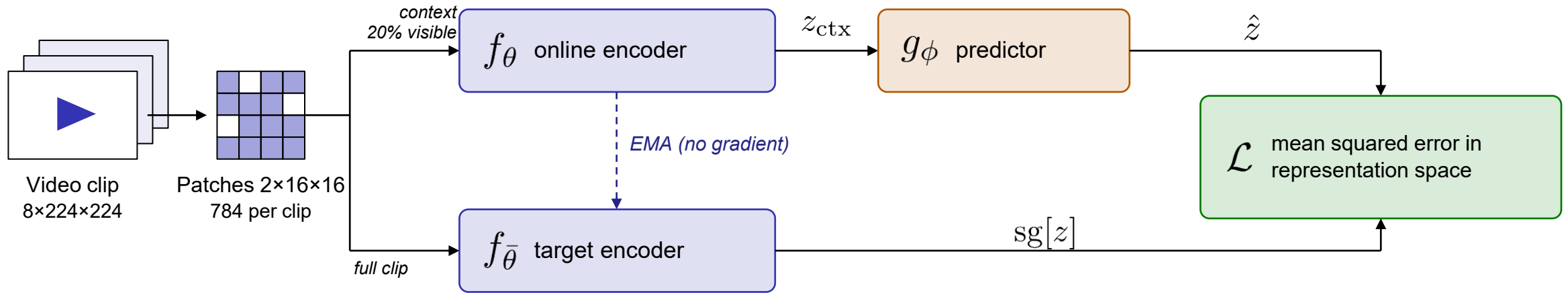
Example: four models classify the same video



Voting: how an ensemble decides

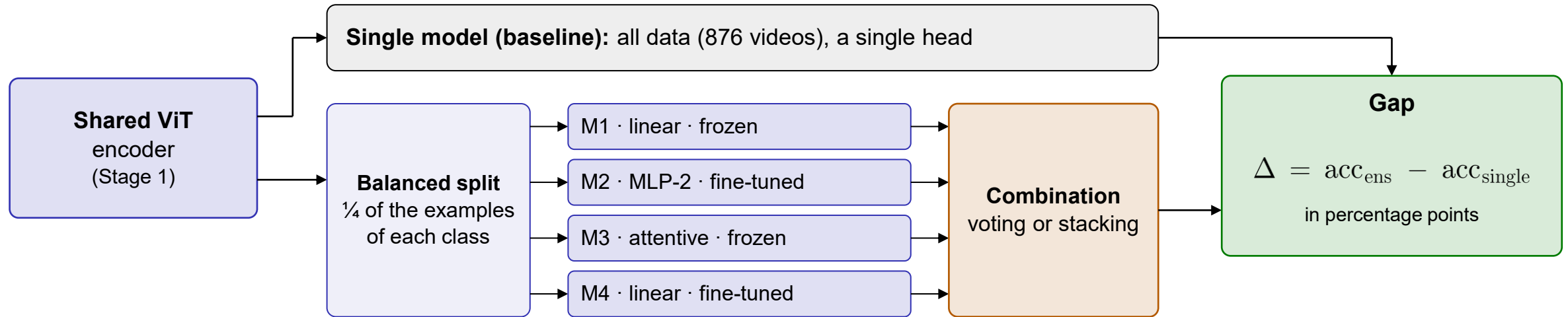
- ▶ Each model gives its opinion and the ensemble decides collectively.
- ▶ It works if the errors differ: correct answers agree and errors cancel out.
- ▶ Hard: the most-voted class wins. Soft: probabilities are averaged. Weighted: better performers count more.
- ▶ Voting applies a fixed rule; stacking learns from data whom to listen to.

Stage 1: “fill in the missing parts” without labels



- ▶ The model sees 20% of the patches (context) and predicts the hidden 80%, using multi-block masking (3 blocks).
- ▶ Prediction happens in representation space, not in pixels.
- ▶ The target encoder is a moving average and receives no gradient: this prevents collapse.
- ▶ Compact ViT: $d = 192$, 8 layers, 8 heads, ≈ 3.4 M parameters; 20 epochs on 3,000 videos.

Stage 2: four different submodels that decide together



- ▶ All four submodels start from the same Stage 1 encoder: only the classification head changes.
- ▶ Each submodel trains on $\frac{1}{4}$ of the 876 videos (≈ 219), with all 30 classes equally represented.
- ▶ Different heads (linear, MLP-2, attentive) and frozen vs. fine-tuned: so each one fails on different cases.
- ▶ The single model sees all 876 videos with one head; the gap Δ measures how much the ensemble gains or loses against it.

Experimental design: one metric and pre-registered verdicts

Kinetics-700-2020 [18] · 30 classes · 1,467 labeled videos · fixed split (seed 1337)

Training · 876

fits M1–M4 and the single model

Validation · 297

selects the configuration and
trains the stacker

Held-out · 294

used only to measure the
gap

Single metric: the gap Δ

$$\Delta = \text{acc}_{\text{ens}} - \text{acc}_{\text{single}}$$

- ▶ In percentage points, on the held-out set.
- ▶ Each video: probabilities averaged over 10 clips.
- ▶ Bootstrap: 10,000 resamples (seed 1337) → 95% CI.

Everything runs on one PC: Intel Core i7 · NVIDIA GTX 1060 (6 GB)

Pre-registered verdicts

SUPPORTED

entire 95% CI above 0

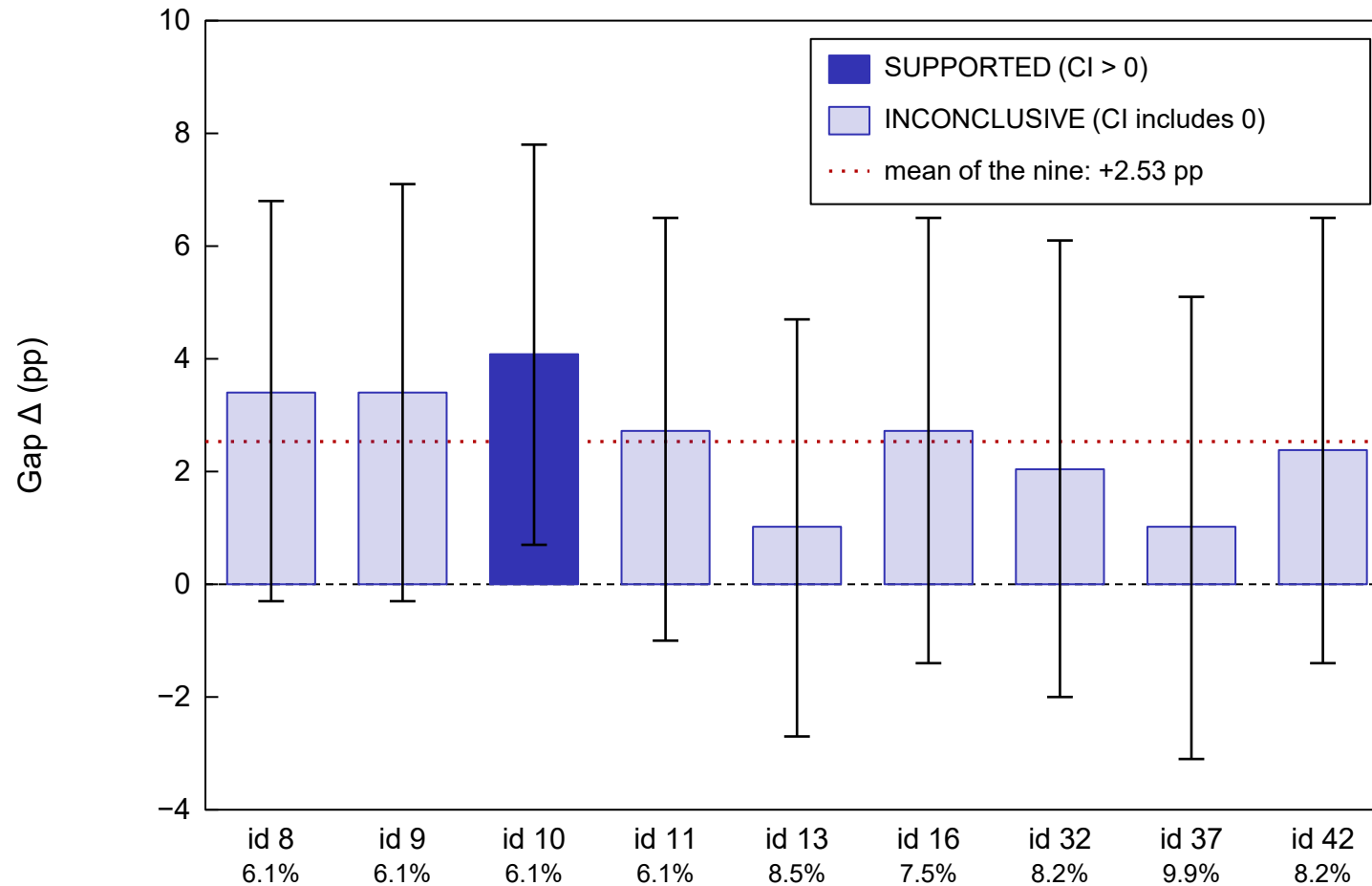
INCONCLUSIVE

$\Delta > 0$, but the CI contains 0

REJECTED

$\Delta \leq 0$

Automated exploration: nine measurements, nine positive gaps



- ▶ An autoresearch-inspired loop proposes configurations, trains the actual models, and logs everything with the fixed metric.
- ▶ **9 of 9 gaps positive**, averaging +2.53 pp.
- ▶ If the true gap were zero, this would be like flipping nine heads in a row:

$$0.5^9 \approx 0.002$$

Main result: only stacking wins with significance

Controlled experiment: same four submodels and same single model; only the combination method changes.

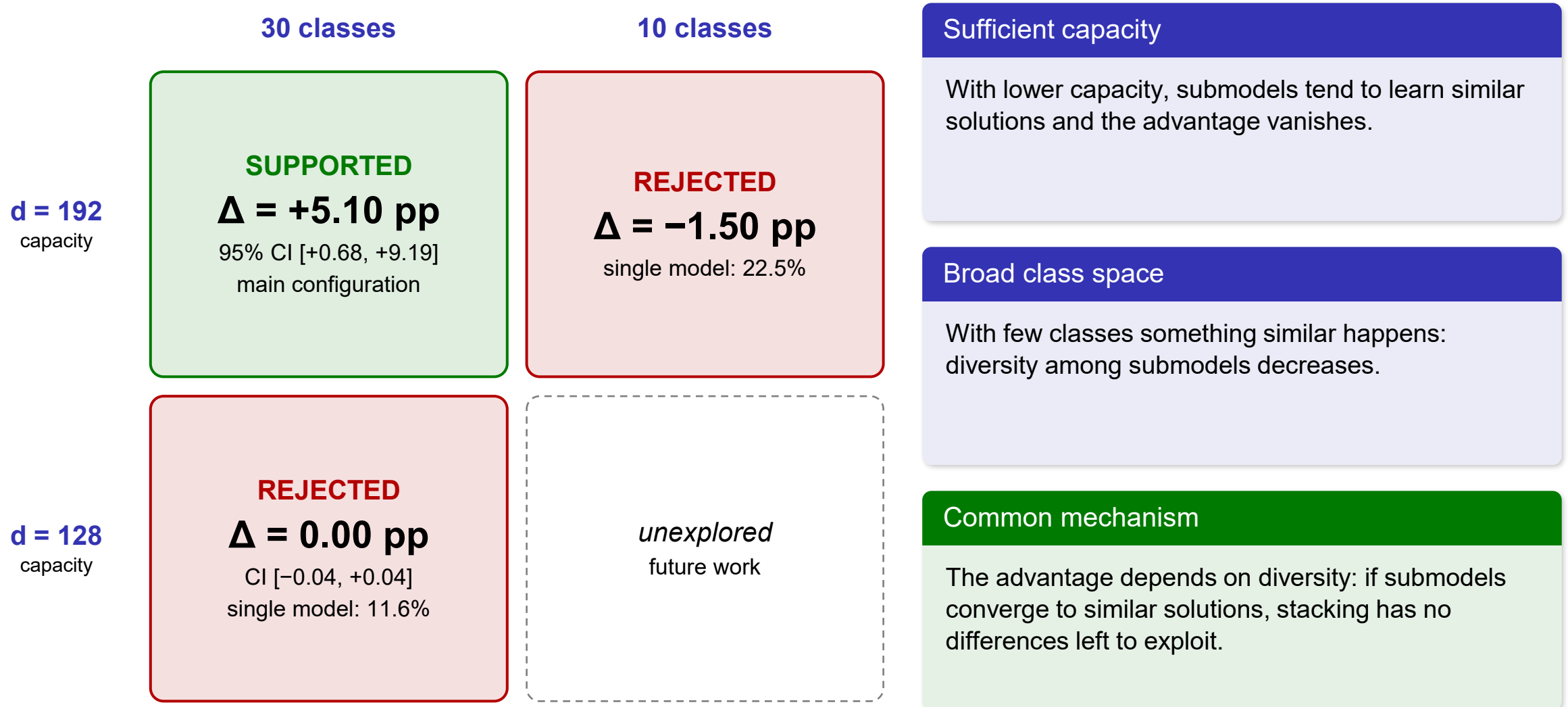
Combination	Accuracy	Δ (pp)	95% CI	Verdict	Δ and 95% CI (pp)
Stacking	11.90%	+5.10	[+0.68, +9.19]	SUPPORTED	
Hard voting	8.84%	+2.04	[-1.70, +5.78]	INCONCLUSIVE	
Soft voting	8.50%	+1.70	[-2.04, +6.12]	INCONCLUSIVE	
Weighted voting	7.48%	+0.68	[-2.72, +4.42]	INCONCLUSIVE	



What the experiment shows

All three voting schemes improve somewhat, but their intervals include zero. Only stacking, which learns whom to listen to for each class, beats the single model significantly (+75% relative accuracy). **Diversity alone is not enough: you have to learn how to combine it.**

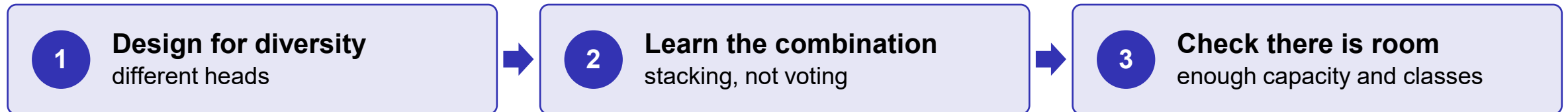
When does it work and when not? The map of conditions



Conclusions: splitting can win, but not for free

Main Result.

The core contribution is not just showing that several small models can beat a large one, but identifying the conditions under which that advantage appears or vanishes.



Future work

Scale up label-free pre-training and study at finer resolution where the advantage appears and vanishes as capacity and number of classes vary.

Thank you!

Questions?

github.com/BHI-Research/pico-jepa-v2